

# Week 4: Fundamentals of Probability and Statistics II

## PLSC 309 Quantitative Political Analysis

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## Final Project: Checkpoint 1

**Due: tomorrow, September 16 at 11:59 p.m. (midnight)**

**Instructions:** Posted on Canvas

## Problem Set 2

- Due: **Next Wednesday (Sep 23rd), 11:59 p.m. (midnight).**
- Already Posted
- It covers Weeks 3-4

## Teaching Observation

- Professor Roseanne McManus will be joining us **next Tuesday** for a teaching observation.

## What Did We Learn Last Time and Last Last Time?

Sep 8th

- Random Variable
- Probability distribution
- Expected value
- Variance and Standard deviation
- Covariance and Correlation

Sep 10th

- Statistical Inference
- Estimand, Estimator, Estimate
- Random sampling
- Independent, Identically Distributed (IID)
- Sampling distribution

# Today's Agenda

- Properties of Estimators
- Interval Estimation

Acknowledgement: Some slides in today's lecture are adapted from Scott Cunningham's [GOV 2001 course materials at Harvard](#) and from Brandon Stewart's [SOC 500 course materials at Princeton](#).

## Part 1. Properties of Estimators

# Many Estimators for the Same Estimand

Suppose our goal is to estimate the **population mean**:

$$\mu = \mathbb{E}[X]$$

There are many possible estimators we could construct from a sample  $X_1, \dots, X_n$ .

**Sample mean**

$$\hat{\mu}_1 = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Uses every observation equally.

**First observation**

$$\hat{\mu}_2 = X_1$$

Uses only one observation.

**Weighted mean**

$$\hat{\mu}_3 = \frac{2}{3} \bar{X}_n + \frac{1}{3} X_1$$

Gives some observations more weight.

**Sample median**

$$\hat{\mu}_4 = \text{Median}(X_1, \dots, X_n)$$

Uses the middle of the sample.

**All are functions of the sample. But are they equally good estimators of  $\mu$ ?**

## Which one should we choose?

- Gets the **right answer on average**  $\rightarrow$  *Unbiasedness*
- Does not vary too much from sample to sample  $\rightarrow$  *Precision / Efficiency*
- Gets closer to the true value as  $n$  increases  $\rightarrow$  *Consistency*
- Has a sampling distribution that becomes predictable for large  $n$   $\rightarrow$  *Asymptotic behavior*

# Properties of Estimators

The properties of an estimator describe the characteristics of the sampling distribution.

**Finite-Sample Properties** (apply for any sample size):

- **Unbiasedness**: Is the estimator centered at the true value?  $\mathbb{E}[\hat{\mu}] = \mu$
- **Efficiency**: Does the estimator have relatively small variance?  
 $\text{Var}(\hat{\mu}_1) < \text{Var}(\hat{\mu}_2)$

**Asymptotic Properties** (apply when  $n$  is large):

- **Consistency**: Does the estimator converge to the true parameter value as  $n$  grows?  $\hat{\theta}_n \xrightarrow{P} \theta$
- **Asymptotic Normality**: Does the sampling distribution approach a normal distribution as  $n$  grows?

## Definition

$$\text{Bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta$$

$\mathbb{E}[\hat{\theta}_n]$  is the expected value of the estimator, and  $\theta$  is the estimand.

Unbiasedness means a statistical estimator does not systematically overestimate or underestimate the true population parameter on average.

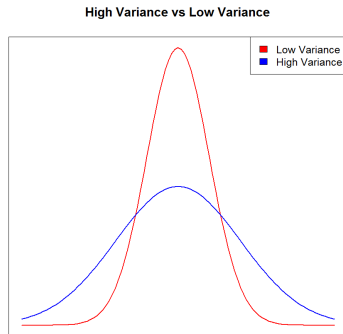
**Unbiased:**  $\text{Bias}(\hat{\theta}_n) = 0$ , i.e.,  $\mathbb{E}[\hat{\theta}_n] = \theta$ .

**Is the sample mean an unbiased estimator of the population mean?**

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \cdot n\mu = \mu$$

**Yes!** On average (over repeated samples),  $\bar{X}_n$  hits the true value.

# Variance of an Estimator



The variance of an estimator measures how much the **estimates** produced by the **estimator** bounce around or spread out across different possible random samples from a population.

# Variance of an Estimator

**Sampling variance:** How spread out is  $\hat{\theta}_n$  around its mean?

$$\text{Var}(\hat{\theta}_n) = \mathbb{E} \left[ \left( \hat{\theta}_n - \mathbb{E}[\hat{\theta}_n] \right)^2 \right]$$

**For the sample mean:**

$$\text{Var}(\bar{X}_n) = \text{Var} \left( \frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{\sigma^2}{n}$$

The variance of the sample mean equals the population variance divided by the sample size.

**Key insight:** Variance shrinks as  $n$  increases—more data means more precision.

# FYI: Why Is $\text{Var}(\bar{X}_n) = \sigma^2/n$ ?

## Variance of the Sample Mean $\bar{X}_n$

**Result:** If  $X_1, X_2, \dots, X_n$  are i.i.d. with  $\mathbb{E}[X_i] = \mu$  and  $\text{Var}(X_i) = \sigma^2$ , then

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

**Derivation:**

|   |                                                                                                                   |                                                                                                                            |
|---|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| ① | $\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$                                     | Start from the definition of the sample mean<br>$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$                                 |
| ② | $= \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right)$                                                         | Use: $\text{Var}(aY) = a^2 \text{Var}(Y)$<br>(a is a constant)                                                             |
| ③ | $= \frac{1}{n^2} \left[ \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j) \right]$ | Use the general formula:<br>$\text{Var}\left(\sum X_i\right) = \sum \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$ |
| ④ | $= \frac{1}{n^2} \left[ \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} 0 \right]$                    | Independence: $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$                                                                    |
| ⑤ | $= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$                                                                    | The second term is 0                                                                                                       |
| ⑥ | $= \frac{1}{n^2} \sum_{i=1}^n \sigma^2$                                                                           | Identically distributed: $\text{Var}(X_i) = \sigma^2$ for all $i$                                                          |
| ⑦ | $= \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}$                                                                | There are $n$ terms in the sum                                                                                             |

**Key Insight:** The variance of the sample mean decreases at rate  $1/n$ .  
More data  $\Rightarrow$  smaller variance  $\Rightarrow$  more precise estimator!

# Efficiency of an Estimator

An estimator is efficient if it has the smallest possible variance (the least amount of spread or variability) (usually among a class of unbiased estimators).

# Standard Error: Precision in Practice

**Standard Error:** The standard deviation of the estimator.

$$SE(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)}$$

**For the sample mean:**  $SE(\bar{X}_n) = \sigma/\sqrt{n}$

**How fast does precision improve?**

- $n = 100$ :  $SE = \sigma/10$
- $n = 10,000$ :  $SE = \sigma/100$

To cut SE in half, you need **4× the sample size** because  $\sqrt{4n} = 2\sqrt{n}$ .

# Estimating the Standard Error

But in practice, the population standard deviation  $\sigma$  is **unknown**.

We estimate  $\sigma$  using the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}$$

Then, the **estimated standard error** is

$$\widehat{SE}(\bar{X}_n) = \frac{s}{\sqrt{n}}$$

**From a single sample, we can calculate both:**

$$\bar{X}_n = \text{Point Estimate} \qquad \frac{s}{\sqrt{n}} = \text{Estimated Standard Error}$$

# Notation Cheat Sheet

## Population

$N$

Population size

$F$

Population distribution

$$\mu = \mathbb{E}[X]$$

Population mean

$$\sigma^2 = \text{Var}(X)$$

Population variance

$\sigma$

Population standard deviation

$p$

Population proportion

## Sample

$n$

Sample size

$$X_1, \dots, X_n$$

Random sample

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Sample mean

$s^2$

Sample variance

$s$

Sample standard deviation

$\hat{p}$

Sample proportion

## Estimation

$\theta$

Estimand (true value)

$\hat{\theta}_n$

Estimator (random variable)

$\hat{\theta}_n^{obs}$

Estimate (realized value)

$\text{Var}(\hat{\theta}_n)$

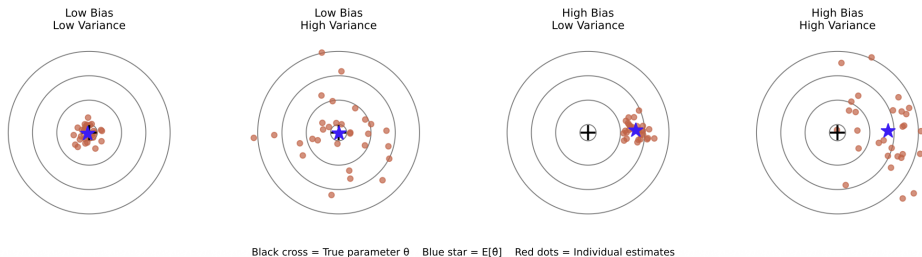
Sampling variance

$SE(\hat{\theta}_n)$

Standard error

# The Bias-Variance Trade-Off

Which matters more—bias or variance?



Neither alone. We care about the overall accuracy of the estimator (Mean squared error).

# Mean Squared Error (MSE)

## Definition

$$\text{MSE}(\hat{\theta}_n) = \mathbb{E} [(\hat{\theta}_n - \theta)^2]$$

MSE measures the average squared difference between the estimated values and the true value.

**Why does  $\text{MSE} = \text{Bias}^2 + \text{Variance}$ ?**

Add and subtract  $\mathbb{E}[\hat{\theta}_n]$ :

$$\begin{aligned}\text{MSE}(\hat{\theta}_n) &= \mathbb{E} [(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n] + \mathbb{E}[\hat{\theta}_n] - \theta)^2] \\ &= \underbrace{\mathbb{E} [(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2]}_{\text{Var}(\hat{\theta}_n)} + \underbrace{(\mathbb{E}[\hat{\theta}_n] - \theta)^2}_{\text{Bias}(\hat{\theta}_n)^2}\end{aligned}$$

Therefore,

$$\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n) + \text{Bias}(\hat{\theta}_n)^2$$

**If the estimator is unbiased:**  $\text{Bias}(\hat{\theta}_n) = 0$ , so  $\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n)$ .

# MSE = Variance + Bias<sup>2</sup>

Start with the definition:

$$\text{MSE}(\hat{\theta}_n) = E[(\hat{\theta}_n - \theta)^2].$$

Add and subtract  $E[\hat{\theta}_n]$ :

$$= E \left[ \{ (\hat{\theta}_n - E[\hat{\theta}_n]) + (E[\hat{\theta}_n] - \theta) \}^2 \right].$$

Expanding:

$$\begin{aligned} &= E[(\hat{\theta}_n - E[\hat{\theta}_n])^2] + (E[\hat{\theta}_n] - \theta)^2 \\ &\quad + 2(E[\hat{\theta}_n] - \theta) \underbrace{E[\hat{\theta}_n - E[\hat{\theta}_n]]}_{=0}. \end{aligned}$$

Therefore,

$$\boxed{\text{MSE}(\hat{\theta}_n) = \underbrace{\text{Var}(\hat{\theta}_n)}_{\text{Variance}} + \underbrace{\text{Bias}(\hat{\theta}_n)^2}_{\text{Bias}^2}}$$

# Why does MSE matter?

- An unbiased estimator can still have high MSE if its variance is large.
- A slightly biased estimator can have lower MSE if it substantially reduces variance.
- This tension is the **bias–variance trade-off**.

MSE measures the average of the squares of the errors between predicted and actual values, making it a critical tool for training and evaluating models. It is a fundamental metric in regression and machine learning.

# Central Limit Theorem: The Intuition

Suppose we repeatedly draw random samples of size  $n$  from a population with mean  $\mu$  and variance  $\sigma^2$ .

Each sample produces a different sample mean:

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots$$

The **Central Limit Theorem (CLT)** tells us that, as  $n$  becomes large, the sampling distribution of  $\bar{X}$  becomes approximately **normal**:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

**Even if the population itself is not normally distributed!**

- Center:  $\mathbb{E}[\bar{X}] = \mu$
- Spread:  $SE(\bar{X}) = \sigma/\sqrt{n}$
- Shape: approximately normal for sufficiently large  $n$

# Why Does the Central Limit Theorem Matter?

The CLT gives us an approximate sampling distribution for the sample mean:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Standardizing:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$$

This allows us to use known probabilities from the standard normal distribution. Therefore, we can quantify the uncertainty around our estimate.

**CLT  $\longrightarrow$  Normal Sampling Distribution  $\longrightarrow$  Confidence Intervals**

# Why Does $Z \sim N(0, 1)$ ?

Suppose  $X \sim N(\mu, \sigma^2)$ . Standardize  $X$ :

$$Z = \frac{X - \mu}{\sigma}.$$

- Subtracting  $\mu$  shifts the center to **0**.
- Dividing by  $\sigma$  rescales the standard deviation to **1**.

$$E(Z) = \frac{E(X) - \mu}{\sigma} = 0, \quad \text{Var}(Z) = \frac{\text{Var}(X)}{\sigma^2} = 1.$$

Because a linear transformation of a normal random variable is still normal,

$$\boxed{Z \sim N(0, 1)}.$$

Z score tells how many standard deviations a specific data point is above or below the mean of a dataset.

## Part 2. Interval Estimation

# From Point Estimation to Interval Estimation

Recall our Pennsylvania polling example:

$$\hat{p} = 0.462$$

This is our **point estimate** of the true population support  $p$ .

But should we believe that

$$p = 0.462 \quad \text{exactly?}$$

# From Point Estimation to Interval Estimation

Recall our Pennsylvania polling example:

$$\hat{p} = 0.462$$

This is our **point estimate** of the true population support  $p$ .

But should we believe that

$$p = 0.462 \quad \text{exactly?}$$

**No.** A different random sample would give a different  $\hat{p}$ .

The point estimate alone does not tell us how **uncertain** our estimate is.

## The Goal

Instead of reporting a single number, construct a **range of plausible values** for the unknown population parameter.

# What Is an Interval Estimate?

A **point estimate** gives us one best guess:  $\hat{\theta}_n$ .

An **interval estimate** gives us a range around that estimate:

$$\text{Point Estimate} \pm \text{Margin of Error}$$

For many estimators:

$$\hat{\theta}_n \pm \underbrace{z^*}_{\text{Critical Value}} \times \underbrace{SE(\hat{\theta}_n)}_{\text{Standard Error}}$$

**Two ingredients determine the width:**

- **Standard error:** How much does the estimator vary across samples?
- **Critical value  $z^*$ :** How far we move from the point estimate?

# From the Sampling Distribution to an Interval

Suppose the sampling distribution is approximately normal:

$$\hat{\theta}_n \approx N\left(\theta, SE(\hat{\theta}_n)^2\right)$$

Standardizing gives:

$$Z = \frac{\hat{\theta}_n - \theta}{SE(\hat{\theta}_n)} \approx N(0, 1)$$

**For a 95% confidence interval, we want the middle 95% of this distribution.**

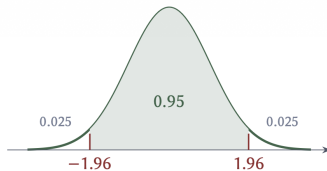
$$P(-1.96 \leq Z \leq 1.96) \approx 0.95$$

The remaining 5% is split equally between the two tails:

$$2.5\% \quad | \quad \mathbf{95\%} \quad | \quad 2.5\%$$

# Why 1.96?

For a 95% confidence interval, we want to capture the **middle 95%** of a standard normal distribution.



*Since  $\hat{Z}_n \approx N(0, 1)$  by the CLT, this 95% area applies to our quantity directly:*

$$\mathbb{P}\left(-1.96 < \frac{\bar{Y} - \mu}{\hat{\sigma}/\sqrt{n}} < 1.96\right) \approx 0.95$$

**Note:** A **normal distribution** is a symmetric, bell-shaped distribution characterized by its mean and standard deviation. The **standard normal distribution** is a normal distribution with mean 0 and standard deviation 1. We use  $Z$  to denote a random variable following this distribution:  $Z \sim N(0, 1)$ .

# Where Does 1.96 Come From?

$$Z \sim N(0, 1)$$

The remaining 5% is split equally between the two tails:

$$\frac{1 - 0.95}{2} = 0.025$$

So the upper cutoff has cumulative probability:

$$1 - 0.025 = 0.975$$

In R:

```
qnorm(0.975, mean = 0, sd = 1, lower.tail = TRUE)
```

```
## [1] 1.959964
```

# Different Confidence Levels

Different confidence levels require different critical values:

| Confidence Level | Critical Value | Interval                   |
|------------------|----------------|----------------------------|
| 90%              | 1.645          | $\hat{\theta} \pm 1.645SE$ |
| 95%              | 1.960          | $\hat{\theta} \pm 1.960SE$ |
| 99%              | 2.576          | $\hat{\theta} \pm 2.576SE$ |

Higher confidence  $\Rightarrow$  larger critical value  $\Rightarrow$  wider interval

**Trade-off:** More confidence requires less precision.

# Different Critical Values

```
qnorm(0.95, mean = 0, sd = 1, lower.tail = TRUE)
```

```
## [1] 1.644854
```

```
qnorm(0.05, mean = 0, sd = 1, lower.tail = TRUE)
```

```
## [1] -1.644854
```

```
qnorm(0.995, mean = 0, sd = 1, lower.tail = TRUE)
```

```
## [1] 2.575829
```

```
qnorm(0.005, mean = 0, sd = 1, lower.tail = TRUE)
```

```
## [1] -2.575829
```

## Example: Pennsylvania Poll

Suppose we survey  $n = 1,000$  Pennsylvania voters:

$$\hat{p} = 0.462$$

For a sample proportion:

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.462(0.538)}{1000}} \approx 0.0158$$

The 95% confidence interval is:

$$\hat{p} \pm 1.96SE(\hat{p})$$

$$0.462 \pm 1.96(0.0158)$$

$$[0.431, 0.493]$$

# Confidence Intervals from Repeated Samples

Suppose the true proportion of Pennsylvania voters who support the candidate is

$$p = 0.40.$$

We repeatedly draw random samples of  $n = 1,000$  voters from this population.

For each sample, we calculate:

Point Estimate  $\longrightarrow$  Standard Error  $\longrightarrow$  95% Confidence Interval.

We repeat this process **100 times**.

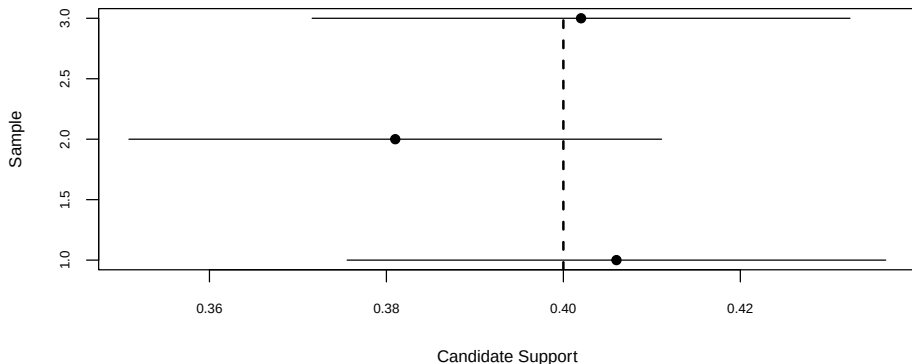
# Confidence Intervals from Repeated Samples

Let's start with just **three random samples**.

Each dot represents the **point estimate**  $\hat{p}$ , and each horizontal line represents its **95% confidence interval**.

The dashed vertical line represents the **true population value**,  $p = 0.40$ .

95% Confidence Intervals from Repeated Samples (3 times)

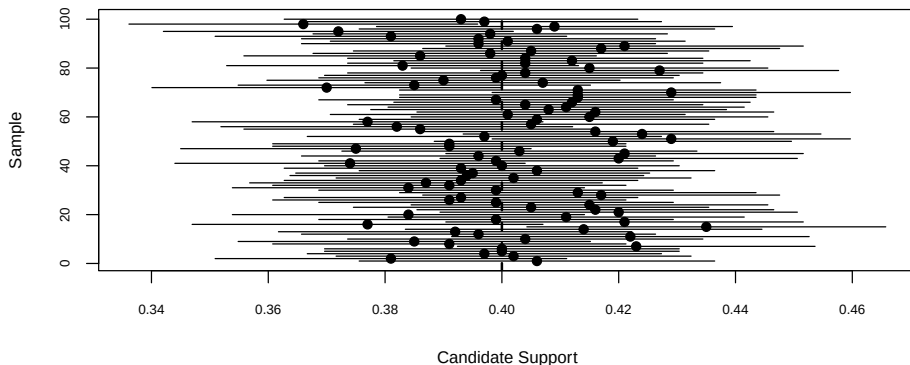


# What If We Repeat This 100 Times?

Now imagine repeating exactly the same procedure **100 times**.

Each random sample produces a different point estimate and therefore a different confidence interval.

95% Confidence Intervals from Repeated Samples (100 times)



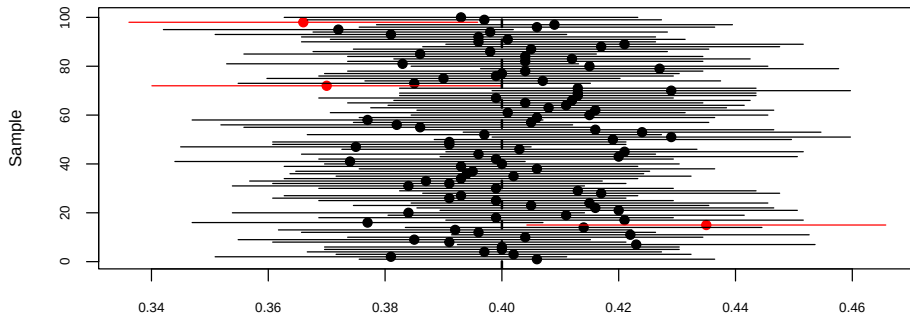
# Do All Confidence Intervals Contain the Truth?

No. Some confidence intervals **miss the true population value**.

In the figure below:

- Black intervals **contain** the true value  $p = 0.40$ .
- Red intervals **do not contain** the true value.

95% Confidence Intervals from Repeated Samples (100 times)



# How to Interpret a 95% Confidence Interval

## Correct

- In repeated sampling, **95% of confidence intervals constructed this way contain the true parameter  $\mu$ .**

## Incorrect

- “There is a 95% probability that  $\mu$  lies within this interval.”
- “95% of the observations in the population lie within this interval.”
- “95% of sample means will lie within this interval.”

**Why?**  $\mu$  is **fixed**. Across repeated samples, the confidence interval changes—not  $\mu$ .

# Key Takeaways

- Properties of Estimator
- Unbiasedness
- Efficiency
- Variance of the Sample Mean
- Standard Error
- Confidence Interval